

UNIT-II

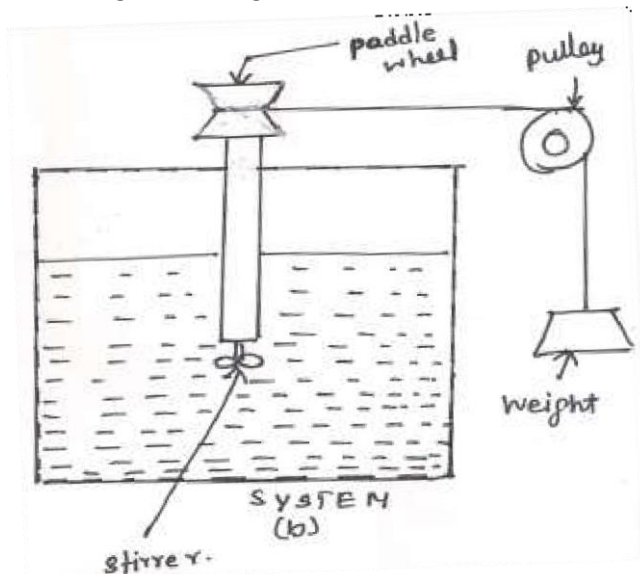
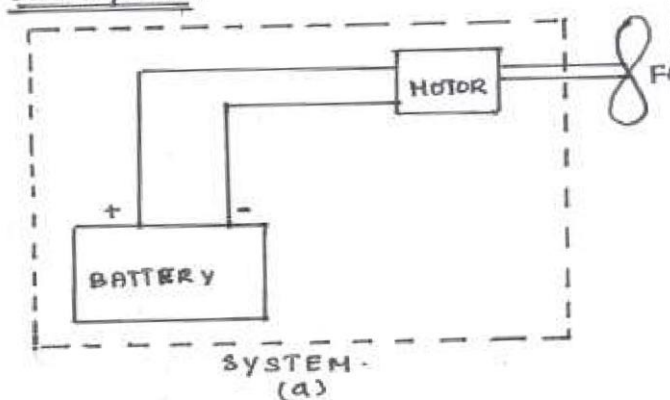
OBJECTIVE:

To familiarize the students with the topics of energy and Entropy and know learn about the 1st Law & 2nd law of thermodynamics and its applications.

THERMODYNAMICS ANALYSIS OF A SYSTEMS

Work is defined as energy in transition between system and surroundings, without transfer of mass. Work is said to be done by a system on its surroundings if the sole effect external to the system could be the raising of a weight.

Examples:



⇒ Let us consider a system which includes a battery and a motor as shown in figure (a). The motor drives a fan (b) the effect of work transfer from the system is driving the fan.

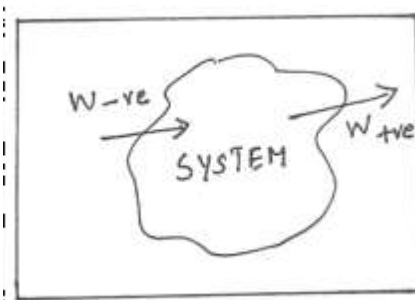
Work is done by the system taken as positive quantity.

Let us consider a system which includes a paddle wheel with a stirrer. To raise or lower the weight, the work is given to the system.

Work is done on the system taken as negative quantity.

SIGN CONVENTION FOR WORK:

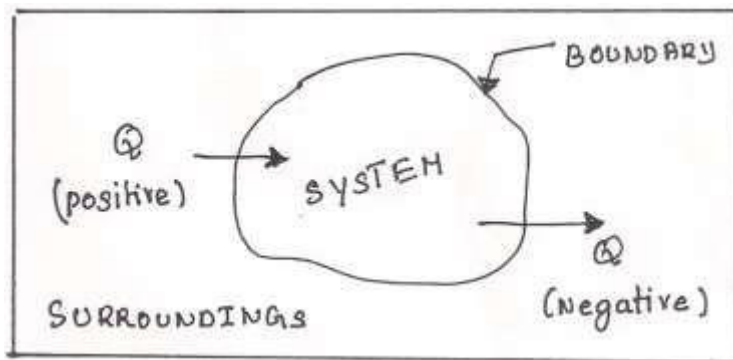
Work done by the system on its surroundings is considered as positive. When work is done on the system as negative.



HEAT TRANSFER:

It is defined as the form of energy. (i.e) transfer between two systems (or a system and its surroundings) by virtue of temperature difference. The transfer of heat into the system is referred as heat addition and transfer of heat out of the stem as heat rejection.

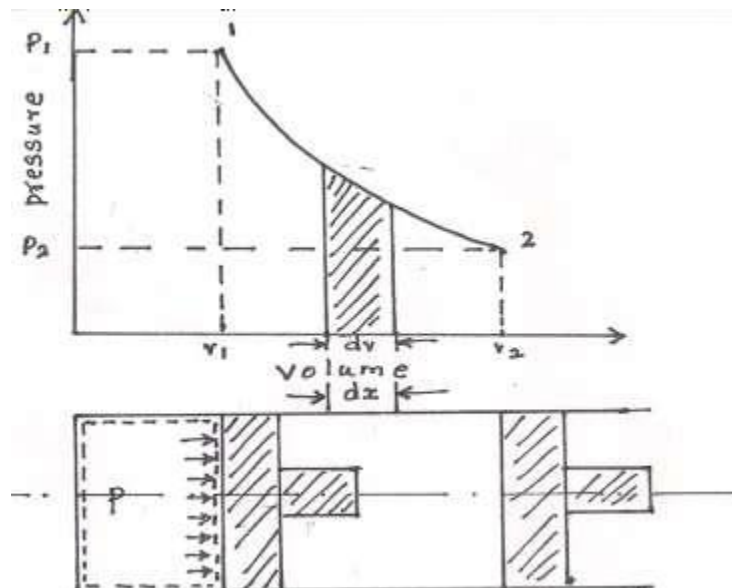
Heat is energy in transition. It is recognized only as it crosses the boundary of the system. The term heat is denoted by ϕ . Heat simply means Heat transfer. Heat flow into the system is taken as positive and heat flow out of the system is taken as negative.



Heat addition ϕ (+ve)

Heat rejection ϕ (-ve)

DISPLACEMENT WORK – PDV WORK:



Displacement Work – Pdv Work:-

Consider a gas enclosed in a piston and cylinder arrangement as shown in figure. The system is in thermodynamic equilibrium in initial position (p_1V_1) and the piston moves to final position, which is also a thermodynamic equilibrium state specified by P_2V_2 .

$$dw = \text{force} \times \text{distance}$$

$$= F \times dx$$

$$\text{But } F = P \times A$$

$$\Rightarrow = PA \times dx$$

$$\text{But } dv = A dx$$

For any process changing, cooling, expansion, compression

$$Dw = p dv$$

$\Rightarrow$ Work done is

$$\int_1^2 dw = \int_1^2 p dv$$

$$W_{1-2} = \int_1^2 p dv$$

Area under the curve in pv-diagram gives the work done under the process.

POINT AND PATH FUNCTIONS:

The quantities pressure, volume, heat, work etc... can be grouped under two classes namely point and path functions.

POINT FUNCTION:

Thermodynamic properties are point function. Since for a given state there is a definite value for each property.

They depend only on the state and not on how the system reaches that state. They have exact differentials.

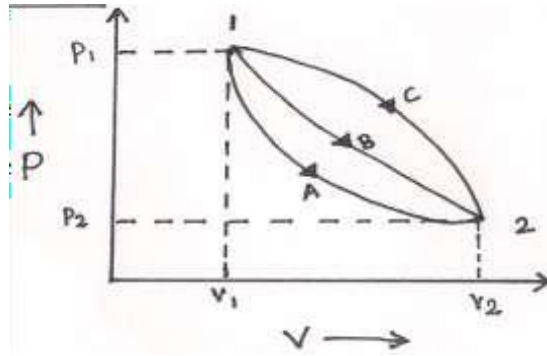
A small change in Volume is represented by dV and the total change in volume during a process

$$\int dv = V_2 - V_1$$

For a cyclic process, the initial and final states are same and hence, the change in any property is zero is

$$\oint dp = 0, \oint dv = 0, \oint dT = 0$$

PATH FUNCTION:



Path Function

Heat and work are path functions. The system from which state 1 to state 2 along many quasi – static process such as A, B and C. Since the curve represents the work for each process.

The amount of work involved in each case is not a function of end state of the process and it depends on the path of the system follows from state 1 to state 2. For this reason, work is called path function and dw is an in – exact differential

$$\int dw \neq w_2 - w_1 \implies \int dw = W_{1-2}$$

FIRST LAW OF THERMODYNAMICS:

First law of thermodynamics states that, “when a system undergoes a cyclic process, then the net heat transfer is equal to the net work transfer”.

Mathematically,

$$\oint_{sw} = \oint \theta Q$$

(or)

$$\oint_{sw} - \oint \theta \phi = 0$$

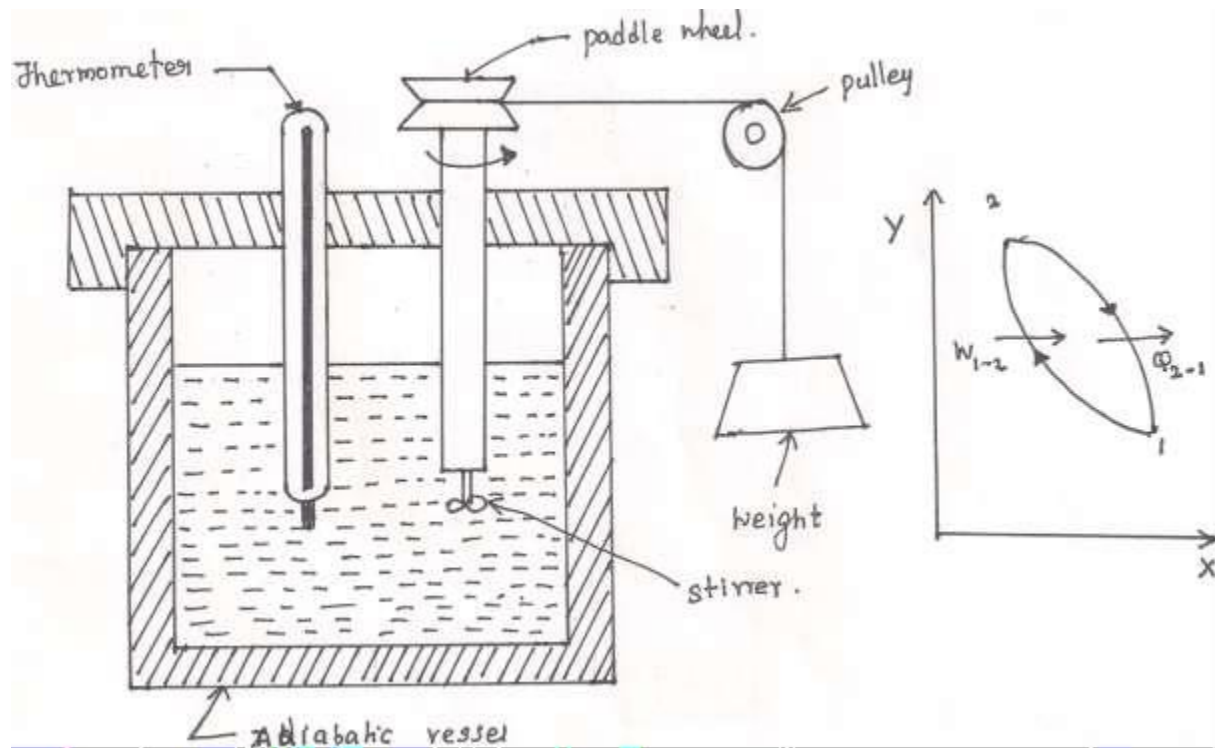
According to law of conservation of energy, “Energy can be neither created nor destroyed but it can transferred to one form to another”. In general for any Thermodynamic systems, the first law of Thermodynamics can be written in the form of following equation.

Heat transfer = Work done + change in Internal Energy.

$$Q = W + \Delta U$$

FIRST LAW FOR A SYSTEM UNDERGOING A CYCLE:

JOULE'S PADDLE WHEEL EXPERIMENT (CLOSED SYSTEM)



Energy which enters the system as heat may leave the system as work or energy which enters the system as work may leave the system as heat.

Let us consider an insulated system which consists of a liquid at a temperature " T_1 " and pressure " P_1 " maintained at atmospheric pressure. A paddle wheel which is placed in the system with weight W . A thermometer is placed in the system to measure the temperature differences.

When the weight falls down, this drives the paddle wheel to pulley. A quantity work can be measured by the fall of weight. During this fall, the paddle wheel stirs the liquid: the temperature of the liquid is increased and taken as " T_2 ". Work transfer: 1-2

Now the insulation is removed such that system and surrounding are interacted by heat transfer till the system returns to its original temperature attaining the condition of Thermal equilibrium with Atmospheric. Heat transfer: 2-1

The system thus executes the cycle which says that an infinite amount of work transfer into a system equal to amount of heat transfer from system. Hence work done during process 1-2 is always proportional to heat transfer during process 2-1

$$W_{1-2} = JQ_{2-1}$$

Where J – Joule equivalent / mechanical equivalent of heat

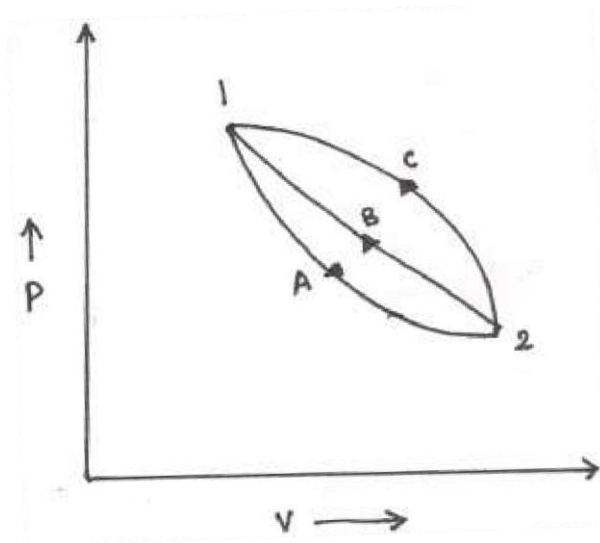
If no. of cycles are involved

$$(\epsilon W)_{\text{cycle}} = (\epsilon \phi)_{\text{cycle}}$$

(or)

$$\oint dw = \oint dQ.$$

ENERGY – THE PROPERTY OF THE SYSTEM:



Consider a system changes from state 1 to state 2 by following path A and returns from state 2 to state 1 by path B.

For path A

$$Q_A = \Delta E_A + W_A$$

$$\implies Q_A - W_A = \Delta E_A$$

For path B

$$Q_B = \Delta E_B + W_B$$

$$\implies Q_B - W_B = \Delta E_B$$

The process A and B together constitute a cycle, for which

$$(\epsilon W)_{\text{cycle}} = (\epsilon \phi)_{\text{cycle}}$$

$$W_A + W_B = \phi_A + \phi_B$$

$$\implies W_A - \phi_A = \phi_B - W_B$$

$$W_A - \phi_A = -[W_B - \phi_B]$$

From 1 & 2

$$\Delta E_A = -\Delta E_B$$

Similarly had the system returned from state 2 to state 1 by following the path C instead of path B.

For path A,

$$\phi_A = \Delta EA + W_A$$

$$\phi_A - W_A = \Delta EA$$

For path C

$$\phi_C = \Delta EC + W_C$$

$$\phi_C - W_C = \Delta EC$$

The process A and C together constitute a cycle, for which

$$(\sum W)_{cycle} = (\sum \phi)_{cycle}$$

$$W_A + W_C = \phi_A + \phi_C \quad \text{From 4 \& 5 from 3 and 6}$$

$$\phi_A - W_A = W_C - \phi_C$$

$$\Delta EA = -EC$$

$$\Delta EB = \Delta EC$$

Therefore it is seen that the change in energy between two states of a system is same, whatever path the system follows undergoing that change of state. Hence it is a point function and the property of the system. The energy E is an extensive property.

ENERGY OF AN ISOLATED SYSTEM:

An Isolated system is one in which there is no intersection of the system with the surrounding.

$$d\phi = 0 :$$

$$dW = 0 :$$

According to first law $dE = 0$; or $E = \text{constant}$ Energy of an isolated system is always constant

SPECIFIC HEAT CAPACITY (C):

The specific heat capacity or heat capacity of a substance is defined as the quantity of heat transfer required to raise or lower the temperature of the unit mass of the substance through one degree.

This is denoted by the symbol 'C'. Its unit is given as J/kgk or kJ / kgk.

SPECIFIC HEAT CAPACITY AT CONSTANT VOLUME [C_v]:-

It is defined as, the quantity of heat transfer required to raise or lower the temperature of unit mass of the substance through one degree when the Volume remain constant. It is denoted by the symbol ' c_v '

When the gas is heated or cooled at constant volume, the heat transfer is given by

$$\phi = mc_v(T_2 - T_1) \text{ kJ}$$

Where, ϕ - Heat transfer (kJ) ; m – mass of the gas (kg)

c_v – Specific heat capacity at constant Volume (kJ / kgK) T_1 –

Initial temperature of the gas (K)

T_2 – Final temperature of the gas (K)

SPECIFIC HEAT CAPACITY AT CONSTANT PRESSURE [C_p]:

It is defined as the quantity of heat transfer required to raise or lower the temperature of unit mass of the substance through one degree when the pressure is kept constant. It is denoted by the symbol c_p .

When the gas is heated or cooled at constant pressure, the heat transfer is given by

$$\phi = mc_p(T_2 - T_1) \text{ J}$$

C_p is always greater than c_v

The ratio of two specific heats (c_p / c_v) remains constant and it is denoted by the symbol γ . It is also called adiabatic index. $\gamma = \frac{c_p}{c_v}$ For air $c_p = 1.005 \text{ kJ / kg.K} = 1.4c_v = 0.718 \text{ kJ / kg.K}$

PERPETUAL MOTION MACHINE (PMM) OF FIRST KIND:

There can be no machine which would continuously supply mechanical work without some other form of energy disappearing simultaneously. Such a fictitious machine is called PMM of first kind.

Thus, a PMM1 is impossible.

INTERNAL ENERGY (U):

Internal energy of a gas is the heat energy stored in the gas at a given temperature. It is denoted by the letter 'U'.

$$U = mc_v t$$

Where

m – Mass of gas (kg)

c_v – Specific heat at constant volume (kJ / kgK)

t – Temperature of the gas (°C)

CHANGE IN INTERNAL ENERGY (ΔU)

When heat is supplied to a certain mass of gas a) the temperature of the gas may increase (or) b) the volume of the gas may increase doing external work (or) c) it may do both.

The increase in heat energy stored in the gas due to rise in temperature is known as increase of internal energy or change in internal energy.

$$\Delta U = mc_v (T_2 - T_1) \text{ kJ}$$

m – Mass of the gas (kg)

c_v – Specific heat capacity at constant Volume (kJ / kgK)

T_1 – Initial temperature of the gas before heating (K)

T_2 – Final Temperature of the gas after heating (K)

FIRST LAW APPLIED TO FLOW PROCESSES

STEADY FLOW PROCESS

Steady flow means that the rates of flow of mass and energy across the control surface are constant.

At the steady state of a system, any thermodynamic property will have a fixed value at a particular location and will not alter with time.

STEADY FLOW ENERGY EQUATION (SFEE):

Consider a thermodynamic system in which a fluid is flowing at a steady state. Fluid enters the system at point 1. And leaves the system at point 2. The space under consideration between entry and exit is known as control volume.

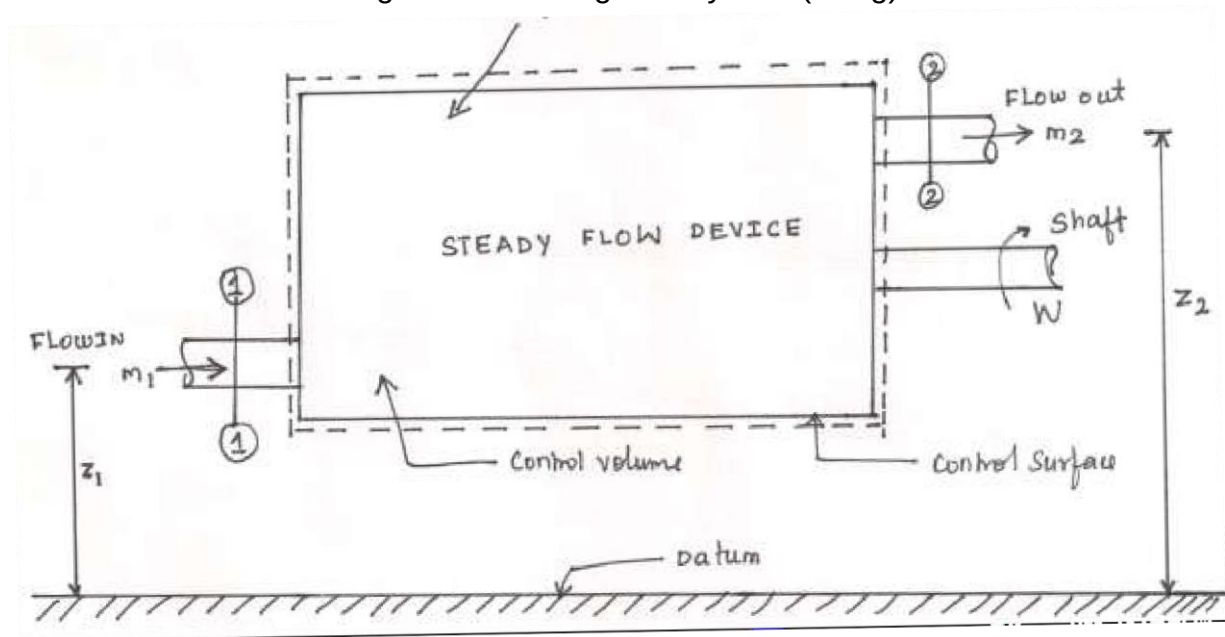
Assumptions made in the system analysis.

1. The mass rate of flow through the control volume is constant.
2. Only potential, kinetic, Internal and flow energies are considered. Other forms of energy are not considered.
3. The interaction between the system and the surroundings are only work and heat. The rate at which work and heat cross the boundary is constant.
4. The state of fluid at any point remains constant at all times.

Let	A+ entry	A+ exit
Intensity of pressure (N/m ²)	P_1	P_2
Internal energy (J / kg)	u_1	u_2
Velocity of flow (m / s)	V_1	V_2
Height above datum(m)	Z_1	Z_2
Specific Volume (m ³ / kg)	V_1	V_2
Enthalpy of working fluid (J /kg)	h_1	h_2
Areas (m ²)	A_1	A_2
Mass (kg)	m_1	m_2

Q – Heat transfer during the flow through the system (J / kg)

W- Work transfer during the flow through the system (J / kg)



MASS BALANCE:

By the conservation of mass, if there is no accumulation of mass within the control volume, the mass flow rate entering must equal the mass flow rate leaving

$$\dot{M}_1 = \dot{M}_2$$

$$\frac{A_1 V_1}{V_1} = \frac{A_2 V_2}{V_2}$$

$$\frac{A_1}{A_2} = \frac{V_2}{V_1} \cdot \frac{V_1}{V_2}$$

$$\frac{\pi d_1^2}{4} V_1 = \frac{\pi d_2^2}{4} V_2$$

$$\frac{d_1^2}{d_2^2} = \sqrt{\frac{V_2}{V_1} \cdot \frac{V_1}{V_2}}$$

ENERGY BALANCE:

By the conservation of energy, the total rate of flow of all energy streams entering the control volume must equal the total rate of flow of all energy streams leaving the control volume.

Total energy entering the system = potential energy + kinetic energy
+ Internal energy + flow energy
+ Heat transfer.

$$\Rightarrow \text{Total energy entering the system} = g Z_1 + \frac{V_1^2}{2} + u_1 + p_1 Q_1 + Q_{\text{in}} \quad \dots \quad 1$$

Total energy leaving the system = potential energy + kinetic energy + Internal energy
+ Flow energy + Work transfer.

$$\text{Total energy leaving the system} = g Z_2 + \frac{V_2^2}{2} + p_2 Q + W \quad \dots 2$$

By law of conservation of energy.

Energy Entering the system = Energy leaving the system From 1 & 2

$$\Rightarrow g Z_1 + \frac{V_1^2}{2} + h_1 + Q = g Z_2 + \frac{V_2^2}{2} + h_2 + W \quad \dots 3$$

$$\Rightarrow Q - W = g (Z_2 - Z_1) + \frac{V_2^2 - V_1^2}{2} + (h_2 - h_1) \quad \dots 4$$

"m" is mass flow rate of the fluid, Eqn. 4 becomes

$$4 \Rightarrow Q - W = m \left[(Z_2 - Z_1) + \frac{V_2^2 - V_1^2}{2} + (h_2 - h_1) \right] \quad \dots 5$$

If h, Q and W are expressed in KJ /kg, Eqn. 5 becomes

$$5 \Rightarrow Q - W = m \left[g \left(\frac{Z_2 - Z_1}{1000} \right) + \frac{V_2^2 - V_1^2}{2000} + (h_2 - h_1) \right]$$

And finally,

$$m \left(\frac{gZ_1}{1000} + \frac{V_1^2}{2000} + h_1 \right) + Q = m \left(\frac{gZ_2}{1000} + \frac{V_2^2}{2000} + h_2 \right) + W$$

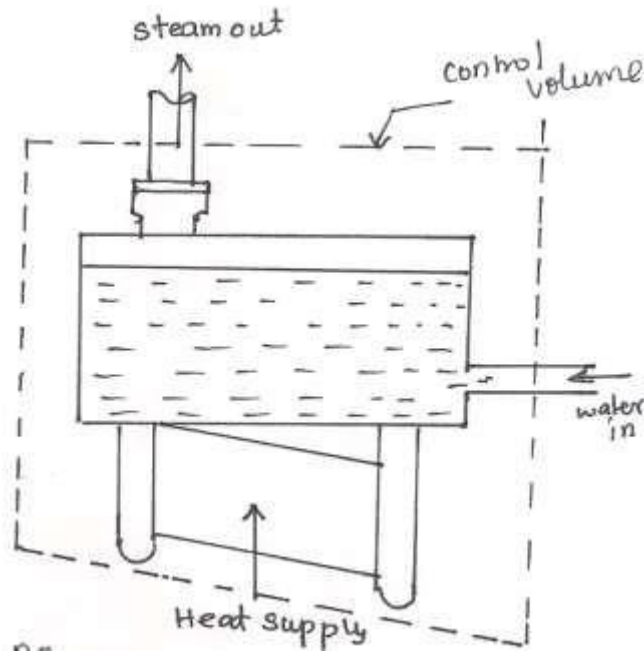
APPLICATIONS OF STEADY FLOW ENERGY EQUATION:-

The steady flow energy equation is applied in

- Steam generators (boilers)
- Steam condensers
- Steam nozzles
- Air compressors
- Steam or gas turbines
- Air heaters.

a) Steam Generator (Or) Boiler:-

Boiler is equipment which generates steam. Water is supplied to the boiler. This water is heated by burning fuel and steam is generated. This steam is supplied to machines for doing work.



Steam Generator

⇒ No mechanical work

$$\text{Is } W = 0$$

⇒ Fluid velocity at the inlet and exit is small. There is no change in kinetic energy

$$\Rightarrow V_1 = V_2$$

Potential energy between inlet and exit is also negligible

$$Z_1 = Z_2$$

For SFEE,

$$Q - W = m \left[g \left(\frac{Z_2 - Z_1}{1000} \right) + \frac{V_2^2 - V_1^2}{2000} + (h_2 - h_1) \right]$$

$$\Rightarrow Q = h_2 - h_1 \text{ j/kg}$$

b) Steam Condenser:-

A steam condenser condenses steam into water. The main function of a condenser is to transfer the heat out of the steady flow system.

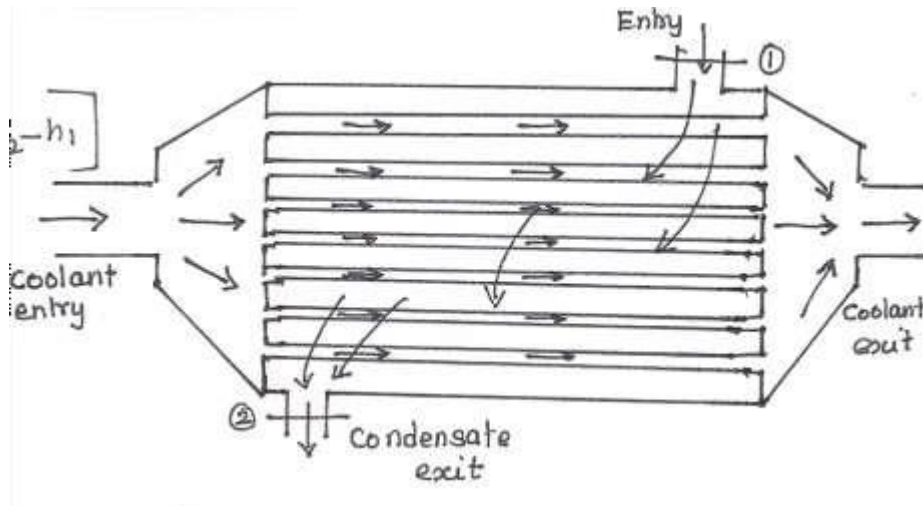
⇒ No work is done $W = 0$

⇒ No change in K.E $V_1 = V_2$

⇒ No change in potential energy $Z_1 = Z_2$

SFEE,

$$Q - W = m \left[g \left(\frac{Z_2 - Z_1}{1000} \right) + \frac{V_2^2 - V_1^2}{2000} + (h_2 - h_1) \right]$$



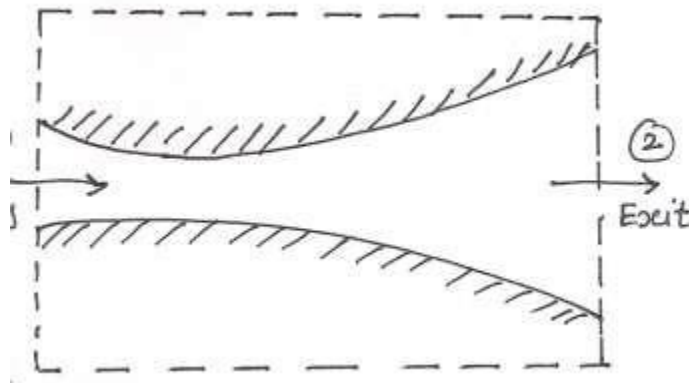
Steam Condenser

Heat transfer,

$$\Rightarrow Q = h_2 - h_1 \text{ j/kg}$$

c) Steam Nozzles:-

Nozzle is a device used for increasing the velocity of flowing fluid at the cost of pressure drop.



Steam Nozzles

$$\Rightarrow Q - W = m \left[g \left(\frac{z_2 - z_1}{1000} \right) + \frac{V_2^2 - V_1^2}{2000} + (h_2 - h_1) \right]$$

$$\frac{V_2^2 - V_1^2}{2000} + (h_2 - h_1) = 0$$

$$\frac{V_2^2 - V_1^2}{2000} = (h_1 - h_2)$$

$$V_2^2 - V_1^2 = 2000(h_1 - h_2)$$

$$V_2^2 = V_1^2 + 2000(h_1 - h_2)$$

$$V_2 = \sqrt{V_1^2 + 2000(h_1 - h_2)} \text{ m/s}$$

h_1, h_2 are in kJ/kg.

d) Air compressors:-

Air compressor is a machine used to produce high pressure air. It takes in atmospheric air and compresses it to a high pressure. It may be broadly classified as i) rotary compressor ii) reciprocating compressors.

Rotary Compressor:-

Rotary compressor has a rotor to develop pressure. In this type of compressors, the rate of flow is very high. Hence this process is treated as reversible adiabatic or isentropic.

SFEE,

$$\text{Work input, } W = h_1 - h_2 \text{ J/kg}$$

Reciprocating Compressor:-

It has a cylinder in which a piston reciprocates. In this type, the rate of flow is comparatively low and large area is in contact with the surroundings. Hence heat transfer is in contact with the surroundings. Hence heat transfer is appreciably more and cannot be neglected.

SFEE,

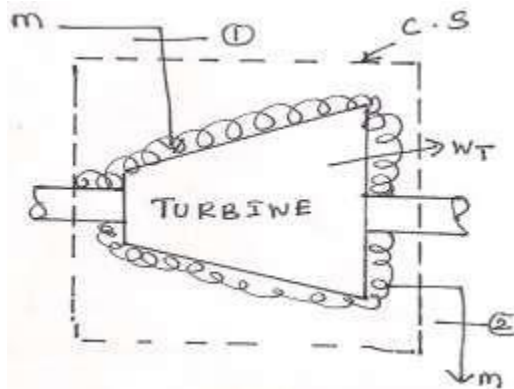
$$\text{Work input, } W = \phi + (h_1 - h_2) \text{ J/kg}$$

e) Steam turbines:-

A steam turbine is a device which converts the energy of steam into mechanical work. Steam is expanded thru a nozzle and a certain amount of head energy is converted into kinetic energy. The steam with high velocity flows over curved blades and its direction of motion is changed. This causes a change of momentum and force thus developed drives the turbine shaft,

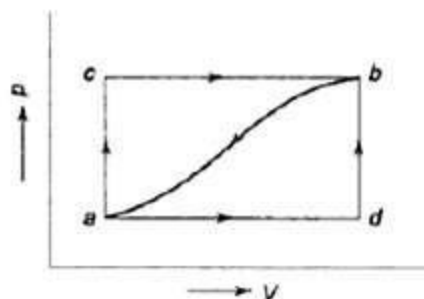
SFEE

Work output, $W = (h_1 - h_2) \text{ J/kg}$



PROBLEMS:

1. When a system is taken from state a to state b , in Fig. Ex. 4.2, along path acb , 84 kJ of heat flow into the system, and the system does 32 kJ of work. (a) How much will the heat that flows into the system along path adb be, if the work done is 10.5 kJ? (b) When the system is returned from b to a along the curved path, the work done on the system is 21 kJ. Does the system absorb or liberate heat, and how much of the heat is absorbed or liberated? (c) If $U_a = 0$ and $U_d = 42$ kJ, find the heat absorbed in the processes ad and db .



Solution

$$Q_{acb} = 84 \text{ kJ}$$

$$W_{acb} = 32 \text{ kJ}$$

We have

$$Q_{acb} = U_b - U_a + W_{acb}$$

$\therefore$

$$U_b - U_a = 84 - 32 = 52 \text{ kJ}$$

Ans.

(a)

$$Q_{adb} = U_b - U_a + W_{adb}$$

$$= 52 + 10.5$$

$$= 62.5 \text{ kJ}$$

Ans.

(b)

$$Q_{b-a} = U_a - U_b + W_{b-a}$$

$$= -52 - 21$$

$$= -73 \text{ kJ}$$

Ans.

2. A piston and cylinder machine contains a fluid system which passes through a complete cycle of four processes. During a cycle, the sum of all heat transfers is -170 kJ. The system completes 100 cycles per min. Complete the following table showing the method for each item, and compute the net rate of work output in kW.

Process	Q (kJ/min)	W (kJ/min)	ΔE (kJ/min)
a-b	0	2,170	—
b-c	21,000	0	—
c-d	-2,100	—	-36,600
d-a	—	—	—

Solution Process *a-b*:

$$Q = \Delta E + W$$

$$0 = \Delta E + 2170$$

$$\therefore \Delta E = -2170 \text{ kJ/min}$$

Process *b-c*:

$$Q = \Delta E + W$$

$$21,000 = \Delta E + 0$$

$$\therefore \Delta E = 21,000 \text{ kJ/min}$$

Process *c-d*:

$$Q = \Delta E + W$$

$$-2100 = -36,600 + W$$

$$\therefore W = 34,500 \text{ kJ/min}$$

Process *d-a*:

$$\sum_{\text{cycle}} Q = -170 \text{ kJ}$$

The system completes 100 cycles/min.

$$\therefore Q_{ab} + Q_{bc} + Q_{cd} + Q_{da} = -17,000 \text{ kJ/min}$$

$$0 + 21,000 - 2,100 + Q_{da} = -17,000$$

$$\therefore Q_{da} = -35,900 \text{ kJ/min}$$

Now $\oint dE = 0$, since cyclic integral of any property is zero.

$$\therefore \Delta E_{a-b} + \Delta E_{b-c} + \Delta E_{c-d} + \Delta E_{d-a} = 0$$

$$-2,170 + 21,000 - 36,600 + \Delta E_{d-a} = 0$$

$$\therefore \Delta E_{d-a} = 17,770 \text{ kJ/min}$$

$$\begin{aligned} \therefore W_{d-a} &= Q_{d-a} - \Delta E_{d-a} \\ &= -35,900 - 17,770 \\ &= -53,670 \text{ kJ/min} \end{aligned}$$

The table becomes

Process	Q (kJ/min)	W (kJ/min)	ΔE (kJ/min)
a-b	0	2,170	-2,170
b-c	21,000	0	21,000
c-d	-2,100	34,500	-36,600
d-a	-35,900	-53,670	17,770

Since

Rate of work output

$$\begin{aligned} \sum_{\text{cycle}} Q &= \sum_{\text{cycle}} W \\ &= -17,000 \text{ kJ/min} \\ &= -283.3 \text{ kW} \end{aligned}$$

Ans.

3. A fluid is confined in a cylinder by a spring-loaded, frictionless piston so that the pressure in the fluid is a linear function of the volume ($p = a + bV$). The internal energy of the fluid is given by the following equation

$$U = 34 + 3.15 pV$$

where U is in kJ, p in kPa, and V in cubic metre. If the fluid changes from an initial state of 170 kPa, 0.03 m^3 to a final state of 400 kPa, 0.06 m^3 , with no work other than that done on the piston, find the direction and magnitude of the work and heat transfer.

Solution The change in the internal energy of the fluid during the process.

$$\begin{aligned} U_2 - U_1 &= 3.15 (p_2 V_2 - p_1 V_1) \\ &= 315 (4 \times 0.06 - 1.7 \times 0.03) \\ &= 315 \times 0.189 = 59.5 \text{ kJ} \end{aligned}$$

$$\begin{aligned} p &= a + bV \\ 170 &= a + b \times 0.03 \\ 400 &= a + b \times 0.06 \end{aligned}$$

From these two equations

$$\begin{aligned} a &= -60 \text{ kN/m}^2 \\ b &= 7667 \text{ kN/m}^5 \end{aligned}$$

Work transfer involved during the process

$$\begin{aligned} W_{1-2} &= \int_{V_1}^{V_2} p dV = \int_{V_1}^{V_2} (a + bV) dV \\ &= a(V_2 - V_1) + b \frac{V_2^2 - V_1^2}{2} \\ &= (V_2 - V_1) \left[a + \frac{b}{2} (V_1 + V_2) \right] \\ &= 0.03 \text{ m}^3 \left[-60 \text{ kN/m}^2 + \frac{7667}{2} \frac{\text{kN}}{\text{m}^5} \times 0.09 \text{ m}^3 \right] \\ &= 8.55 \text{ kJ} \end{aligned}$$

Work is done by the system, the magnitude being 8.55 kJ.

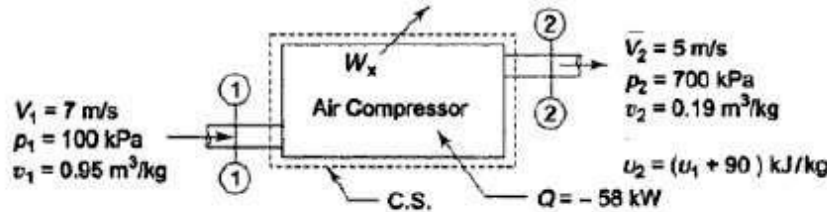
∴ Heat transfer involved is given by

$$\begin{aligned} Q_{1-2} &= U_2 - U_1 + W_{1-2} \\ &= 59.5 + 8.55 \\ &= 68.05 \text{ kJ} \end{aligned}$$

68.05 kJ of heat flow into the system during the process.

4. Air flows steadily at the rate of 0.5 kg/s through an air compressor, entering at 7 m/s velocity, 100 kPa pressure, and 0.95 m³/kg volume, and leaving at 5 m/s, 700 kPa, and 0.19 m³/kg. The internal energy of the air leaving is 90 kJ/kg greater than that of the air entering. Cooling water in the compressor jackets absorbs heat from the air at the rate of 58 kW. (a) Compute the rate of shaft work input to the air in kW. (b) Find the ratio of the inlet pipe diameter to outlet pipe diameter.

Solution Figure shows the details of the problem.



(a) Writing the steady flow energy equation, we have

$$\begin{aligned}
 w \left(u_1 + p_1 v_1 + \frac{V_1^2}{2} + Z_1 g \right) + \frac{dQ}{d\tau} \\
 &= w \left(u_2 + p_2 v_2 + \frac{V_2^2}{2} + Z_2 g \right) + \frac{dW_x}{d\tau} \\
 \therefore \frac{dW_x}{d\tau} &= -w \left[(u_2 - u_1) + (p_2 v_2 - p_1 v_1) + \frac{V_2^2 - V_1^2}{2} + (Z_2 - Z_1)g \right] + \frac{dQ}{d\tau} \\
 \therefore \frac{dW_x}{d\tau} &= -0.5 \frac{\text{kg}}{\text{s}} \left[90 \frac{\text{kJ}}{\text{kg}} + (7 \times 0.19 - 1 \times 0.95) 100 \frac{\text{kJ}}{\text{kg}} \right. \\
 &\quad \left. + \frac{(5^2 - 7^2) \times 10^{-3}}{2} \frac{\text{kJ}}{\text{kg}} + 0 \right] - 58 \text{ kW} \\
 &= -0.5 [90 + 38 - 0.012] \text{ kJ/s} - 58 \text{ kW} \\
 &= -122 \text{ kW}
 \end{aligned}$$

Ans. (a)

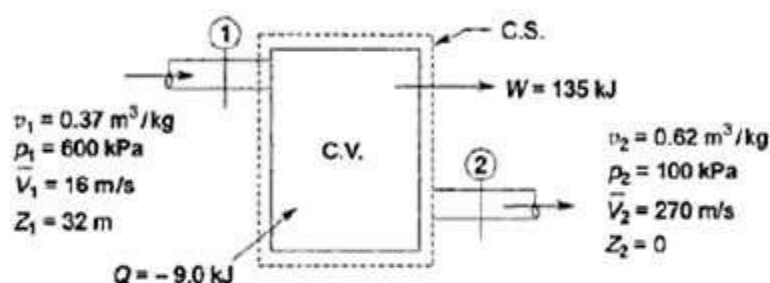
Rate of work input is 122 kW.

(b) From mass balance, we have

$$\begin{aligned}
 w &= \frac{A_1 V_1}{v_1} = \frac{A_2 V_2}{v_2} \\
 \therefore \frac{A_1}{A_2} &= \frac{v_1}{v_2} \cdot \frac{V_2}{V_1} = \frac{0.95}{0.19} \times \frac{5}{7} = 3.57 \\
 \therefore \frac{d_1}{d_2} &= \sqrt{3.57} = 1.89
 \end{aligned}$$

5. In a steady flow apparatus, 135 kJ of work is done by each kg of fluid. The specific volume of the fluid, pressure, and velocity at the inlet are $0.37 \text{ m}^3/\text{kg}$, 600 kPa, and 16 m/s. The inlet is 32 m above the floor, and the discharge pipe is at floor level. The discharge conditions are $0.62 \text{ m}^3/\text{kg}$, 100 kPa, and 270 m/s. The total heat loss between the inlet and discharge is 9 kJ/kg of fluid. In flowing through this apparatus, does the specific internal energy increase or decrease, and by how much?

Solution Writing the steady flow energy equation for the control volume,



$$u_1 + p_1 v_1 + \frac{V_1^2}{2} + Z_1 g + \frac{dQ}{dm} = u_2 + p_2 v_2 + \frac{V_2^2}{2} + Z_2 g + \frac{dW_s}{dm}$$

$$\therefore u_1 - u_2 = (p_2 v_2 - p_1 v_1) + \frac{V_2^2 - V_1^2}{2} + (Z_2 - Z_1)g + \frac{dW_s}{dm} - \frac{dQ}{dm}$$

$$\begin{aligned} &= (1 \times 0.62 - 6 \times 0.37) \times 10^2 + \frac{(270^2 - 16^2) \times 10^{-3}}{2} \\ &\quad + (-32 \times 9.81 \times 10^{-3}) + 135 - (-9.0) \\ &= -160 + 36.45 - 0.314 + 135 + 9 \\ &= 20.136 \text{ kJ/kg} \end{aligned}$$

Specific internal energy decreases by 20.136 kJ.

6. Air at a temperature of 15°C passes through a heat exchanger at a velocity of 30 m/s where its temperature is raised to 800°C . It then enters a turbine with the same velocity of 30 m/s and expands until the temperature falls to 650°C . On leaving the turbine, the air is taken at a velocity of 60 m/s to a nozzle where it expands until the temperature has fallen to 500°C . If the air flow rate is 2 kg/s, calculate (a) the rate of heat transfer to the air in the heat exchanger, (b) the power output from the turbine assuming no heat loss, and (c) the velocity at exit from the nozzle, assuming no heat loss. Take the enthalpy of air as $h = c_p t$, where c_p is the specific heat equal to 1.005 kJ/kg K and t the temperature.

Solution As shown in Fig. writing the S.F.E.E. for the heat exchanger and eliminating the terms not relevant,

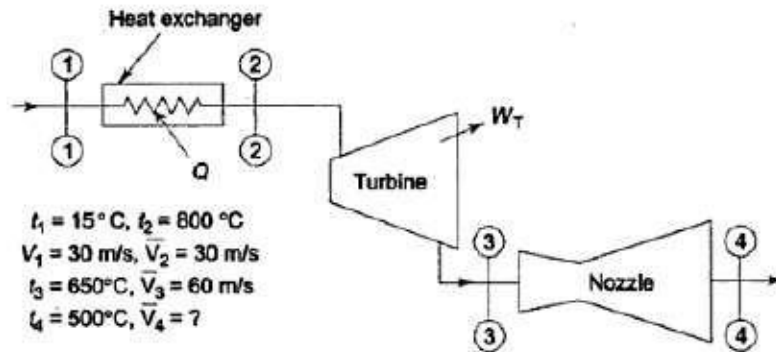
Solution As shown in Fig. writing the S.F.E.E. for the heat exchanger and eliminating the terms not relevant,

$$w \left(h_1 + \frac{V_1^2}{2} + Z_1 g \right) + Q_{1-2} = w \left(h_2 + \frac{V_2^2}{2} + Z_2 g \right) + W_{1-2}$$

$$\therefore wh_1 + Q_{1-2} = wh_2$$

$$\begin{aligned} \therefore Q_{1-2} &= w(h_2 - h_1) = wc_p (t_2 - t_1) \\ &= 2 \times 1.005 (800 - 15) \\ &= 2.01 \times 785 \\ &= 1580 \text{ kJ/s} \end{aligned}$$

Ans. (a).



Energy equation for the turbine gives

$$w \left(\frac{V_2^2}{2} + h_2 \right) = wh_3 + w \frac{V_3^2}{2} + W_T$$

$$\frac{V_2^2 - V_3^2}{2} + (h_2 - h_3) = W_T/w$$

$$\frac{(30^2 - 60^2) \times 10^{-3}}{2} + 1.005 (800 - 650) = W_T/w$$

$$\therefore \frac{W_T}{w} = -1.35 + 150.75$$

$$= 149.4 \text{ kJ/kg}$$

$$\therefore W_T = 149.4 \times 2 \text{ kJ/s}$$

$$= 298.8 \text{ kW}$$

Ans. (b)

Writing the energy equation for the nozzle

$$\frac{V_3^2}{2} + h_3 = \frac{V_4^2}{2} + h_4$$

$$\frac{V_4^2 - V_3^2}{2} = c_p (t_3 - t_4)$$

$$V_4^2 - V_3^2 = 1.005 (650 - 500) \times 2 \times 10^3$$

$$= 301.50 \times 10^3 \text{ m}^2/\text{s}^2$$

$$= 30.51 \times 10^4 \text{ m}^2/\text{s}^2$$

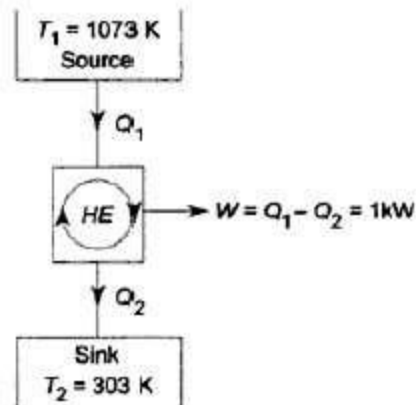
$\therefore$ Velocity at exit from the nozzle

$$V_4 = 554 \text{ m/s}$$

Ans. (c)

7. A cyclic heat engine operates between a source temperature of 800°C and a sink temperature of 30°C . What is the least rate of heat rejection per kW net output of the engine?

Solution For a reversible engine, the rate of heat rejection will be minimum



$$\begin{aligned}\eta_{\max} = \eta_{\text{rev}} &= 1 - \frac{T_2}{T_1} \\ &= 1 - \frac{30 + 273}{800 + 273} \\ &= 1 - 0.282 = 0.718\end{aligned}$$

Now $\frac{W_{\text{net}}}{Q_1} = \eta_{\max} = 0.718$

$\therefore Q_1 = \frac{1}{0.718} = 1.392 \text{ kW}$

Now $Q_2 = Q_1 - W_{\text{net}} = 1.392 - 1$
 $= 0.392 \text{ kW}$

This is the least rate of heat rejection.

8. A reversible heat engine in a satellite operates between a hot reservoir at T_1 and a radiating panel at T_2 . Radiation from the panel is proportional to its area and to T_2^4 . For a given work output and value of T_1 show that the area of the panel will be minimum when $\frac{T_2}{T_1} = 0.75$.

Determine the minimum area of the panel for an output of 1 kW if the constant of proportionality is $5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$ and T_1 is 1000 K.

Solution For the heat engine (Fig. Ex. 6.7), the heat rejected Q_2 to the panel (at T_2) is equal to the energy emitted from the panel to the surroundings by radiation. If A is the area of the panel, $Q_2 \propto AT_2^4$, or $Q_2 = KAT_2^4$, where K is a constant.

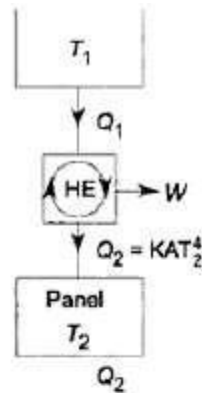


Fig. Ex. 6.7

Now
$$\eta = \frac{W}{Q_1} = \frac{T_1 - T_2}{T_1}$$

or
$$\frac{W}{T_1 - T_2} = \frac{Q_1}{T_1} = \frac{Q_2}{T_2} = \frac{KAT_2^4}{T_2} = KAT_2^3$$

$\therefore A = \frac{W}{KT_2^3 (T_1 - T_2)} = \frac{W}{K(T_1 T_2^3 - T_2^4)}$

For a given W and T_1 , A will be minimum when

$$\frac{dA}{dT_2} = -\frac{W}{K} (3T_1 T_2^2 - 4T_2^3) \cdot (T_1 T_2^3 - T_2^4)^{-2} = 0$$

Since $(T_1 T_2^3 - T_2^4)^{-2} \neq 0$, $3T_1 T_2^2 = 4T_2^3$

$\therefore \frac{T_2}{T_1} = 0.75$ Proved.

$\therefore A_{\min} = \frac{W}{K(0.75)^3 T_1^3 (T_1 - 0.75 T_1)}$

$$= \frac{W}{K \frac{27}{256} T_1^4} = \frac{256 W}{27 K T_1^4}$$

Here $W = 1 \text{ kW}$, $K = 5.67 \times 10^{-8} \text{ W/m}^2 \text{ K}^4$, and $T_1 = 1000 \text{ K}$

$\therefore A_{\min} = \frac{256 \times 1 \text{ kW} \times \text{m}^2 \text{ K}^4}{27 \times 5.67 \times 10^{-8} \text{ W} \times (1000)^4 \text{ K}^4}$

$$= \frac{256 \times 10^3}{27 \times 5.67 \times 10^{-8} \times 10^{12}} \text{ m}^2$$

$$= 0.1672 \text{ m}^2$$

Ans.